

Forder — Book IV

Proposition 19

Formal Reconstruction — Technical Documentation

Jurek Rózański
Military University of Technology, Warsaw, Poland
`jerzy.rozanski@wat.edu.pl`

CGJTEAMLAB

If C, D lie on the same side of AB and the angles ACB and ADB are congruent, then A, B, C, D are concyclic.

1 Introduction

Forder IV.19 is the converse of the same-segment theorem.

In source notation, if C, D lie on the same side of chord AB and

$$\angle ACB \cong \angle ADB,$$

then the four points A, B, C, D are cyclic [1].

The production interface uses a relabelled but equivalent form. The fixed chord is CD ; the points A, B lie on the same side of its carrier; and the equal-angle hypothesis is

$$\angle CAD \cong \angle CBD.$$

The conclusion is

$$A, C, D, B \text{ are concyclic.}$$

This proposition is a genuine converse theorem. The circle containing all four points is not assumed. The proof therefore begins by constructing the circumcircle through three points and then proves that the fourth point must lie on it.

The formal architecture is:

$$\text{circle through } C, D, A \longrightarrow \begin{cases} B \text{ already on the circle,} \\ B \text{ off the circle} \end{cases}$$

and in the second case:

$$\text{line } CB \longrightarrow \begin{cases} \text{secant,} \\ \text{tangent.} \end{cases}$$

The secant branch is itself divided into two ray-order cases. The tangent branch is excluded by IV.18.

The production implementation is

```
CGJteamLab/Proposition4_19.lean
```

and it exposes both a dependency-explicit theorem

```
hilbert_IV19_circle_converse_of_IV16_IV18
```

and the closed public endpoint

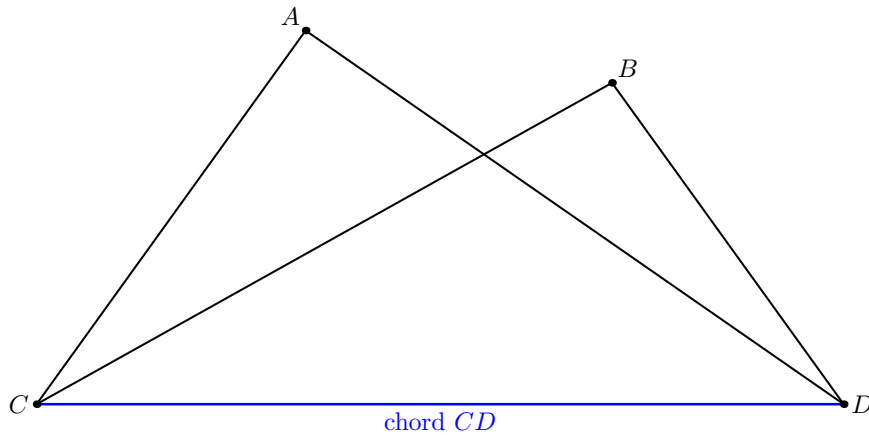
```
hilbert_IV19_circle_converse
```

2 Classical Configuration

Let C, D determine the chord carrier. Assume A, B lie on the same side of that line and

$$\angle CAD \cong \angle CBD.$$

A, B same side of CD



$$\angle CAD \cong \angle CBD$$

Figure 1: Production naming for Forder IV.19. The points A, B lie on the same side of chord CD , and the equal-angle hypothesis is $\angle CAD \cong \angle CBD$. The theorem concludes that A, C, D, B are concyclic.

The source theorem is usually remembered as the converse of “angles in the same segment are equal.” The formal proof, however, cannot simply invoke a uniqueness theorem for circles from an equality of angle measures. It constructs the relevant circle and excludes all ways in which the fourth point could fail to belong to it.

3 Forder’s Statement and Proof

Forder argues as follows.

Because A, C, D are noncollinear, there is a circle through those three points. If B lies on that circle, the theorem is proved.

Suppose B does not lie on it. Then the line through C and B either meets the circle again in a point $E \neq C$, or is tangent to the circle at C .

In the secant case, IV.16 compares the angle at E with the angle at A . According to the order of E relative to B and C , the resulting angles are either congruent or supplementary. Both alternatives contradict elementary triangle angle order unless $E = B$. But $E = B$ would put B on the circle.

In the tangent case, IV.18 gives the tangent–chord angle relation. Combined with the assumed equality of the two inscribed angles, this makes a remote interior angle congruent to an exterior angle of a triangle, again impossible.

Thus the line CB cannot have a second circle point distinct from B and cannot be tangent while B remains off the circle. Therefore B must lie on the constructed circumcircle.

The formal proof preserves this source architecture almost literally, but makes every line-order and side-of-chord distinction explicit.

4 Formal Statement

The proposition interface in `HilbertInterfaceIV.lean` is:

```
def HilbertForderIV19CircleConverse
  [H : HilbertIncidence Geo]
  [_HO : @HilbertOrder Geo H] : Prop :=
  forall C D A B : Geo.Point,
    forall chord : Geo.Line,
      Ne C D ->
      H.OnLine C chord ->
      H.OnLine D chord ->
      HilbertSameSide Geo A B chord ->
      Geo.AngleCongruent
        C A D
        C B D ->
      HilbertConcyclic4 Geo A C D B
```

The hypothesis

```
HilbertSameSide Geo A B chord
```

contains the off-line information for both A and B . In particular, A is off chord CD , so C, D, A form a genuine triangle.

The closed production theorem is:

```
theorem hilbert_IV19_circle_converse
  [H : HilbertIncidence Geo]
  [HE : @HilbertEuclideanPlane Geo H] :
  HilbertForderIV19CircleConverse (Geo := Geo)
```

The production module also exports a dependency-explicit variant

```
hilbert_IV19_circle_converse_of_IV16_IV18
```

which exposes the mathematical fact that the proof requires both halves of IV.16 and IV.18.

5 Constructing the Circumcircle Through Three Points

Since A lies off the carrier CD , the points

$$C, D, A$$

are noncollinear.

The proof invokes the triangle circumcenter theorem

```
hilbert_triangle_circumcenter_exists_XI
```

to construct a point K satisfying

$$KC \cong KD, \quad KC \cong KA.$$

Using C itself as the reference-radius point, the proof obtains circle membership:

$$C, D, A$$

all lie on the circle centered at K .

The theorem now splits on whether B belongs to this circle.

If it does, the target

$$\text{Concyclic4}(A, C, D, B)$$

is immediate.

All remaining work occurs under the contradiction hypothesis that B is off the constructed circle.

6 Preparing the Carrier CB

Assume B is not on the constructed circle.

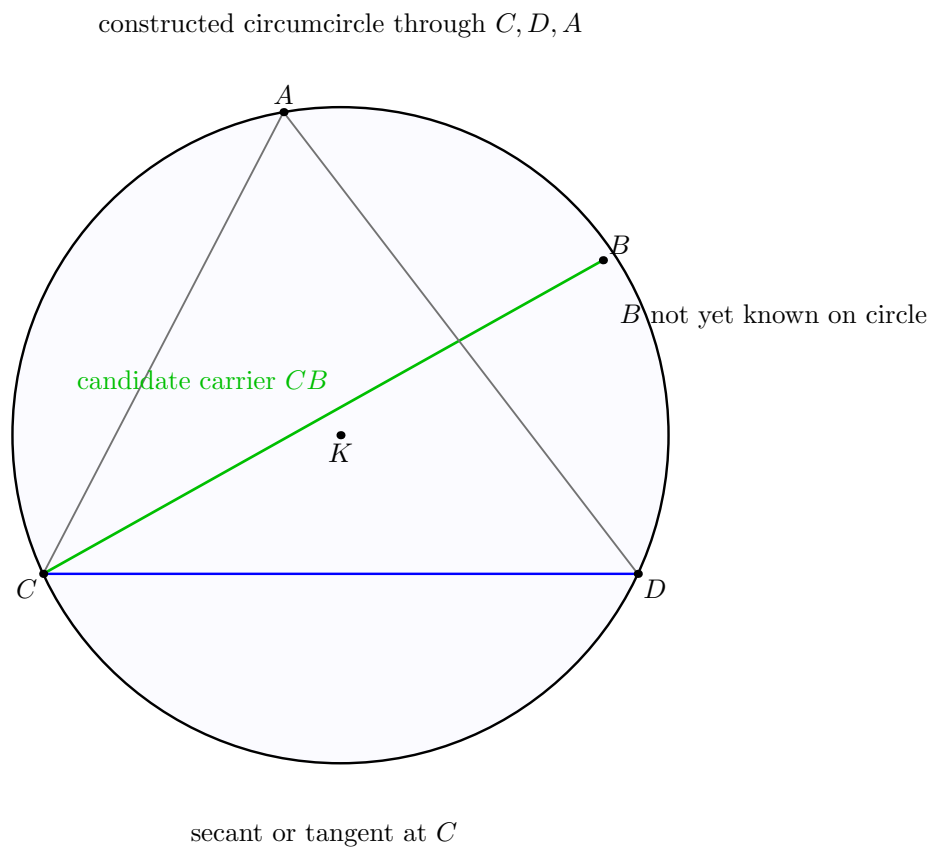


Figure 2: The converse proof starts by constructing the circumcircle through C, D, A . The fourth point B is not initially assumed to lie on this circle. If B is off it, the carrier CB is analyzed as secant or tangent.

Because B lies on the same side of chord CD as A , it is off that chord. In particular

$$C \neq B.$$

Let ℓ_{CB} be the carrier through C, B .

The proof also shows

$$D \notin \ell_{CB}.$$

Otherwise C, D, B would be collinear, forcing B onto chord CD , contrary to the same-side hypothesis.

The circle centre is distinct from C , so the general theorem

```
hilbert_circle_line_second_or_tangent
```

applies to the line ℓ_{CB} through circle point C .

It returns exactly two alternatives:

1. there exists a second circle point $E \neq C$ on ℓ_{CB} ;
2. ℓ_{CB} is tangent to the circle at C .

This is the main structural dichotomy of IV.19.

7 Secant Branch: Order Normalization

Suppose $E \neq C$ is the second circle point on line CB .

Since B was assumed off the circle,

$$E \neq B.$$

The three points C, B, E are distinct and collinear. The reusable theorem

```
hilbert_secant_ray_order_split
```

reduces all possible linear orders to two geometrically relevant cases:

$$\text{SameRay}(C, B, E)$$

or

$$B - C - E.$$

This semantic production theorem was originally discovered inside the IV.19 development, but it now belongs to `HilbertInterfaceIV`. Its promotion is essential to the acyclic Book IV architecture, because IV.17 also needs the same general ray-order fact.

8 Same-Ray Secant Branch

Assume

$$\text{SameRay}(C, B, E).$$

The point E lies on the same circle as C, D, A .

Because A, B lie on the same side of chord CD , and E lies on the same ray CB as B , the relevant IV.16 same-side configuration gives

$$\angle CAD \cong \angle CED.$$

The IV.19 hypothesis is

$$\angle CAD \cong \angle CBD.$$

Hence

$$\angle CED \cong \angle CBD.$$

But B, E lie on the same ray from C . This is exactly the setting of the reusable secant-rigidity theorem

`hilbert_equal_angles_same_secant_unique`

which says that if two points on the same ray from a fixed origin subtend the same angle against a point off the carrier, then the two points coincide.

Therefore

$$E = B.$$

This contradicts $E \neq B$, which came from the assumption that B is off the circle.

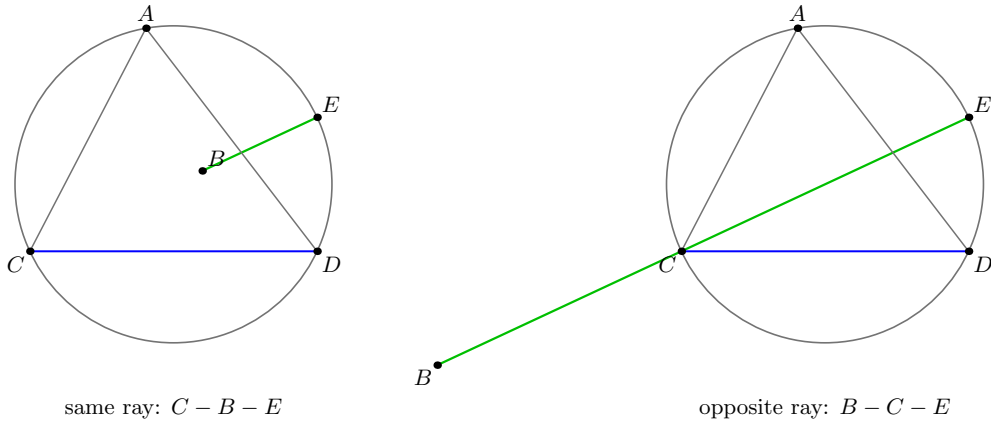


Figure 3: The two secant orders in IV.19. Left: E lies on the same ray CB as B , and equal-angle secant rigidity forces $E = B$. Right: $B - C - E$; then B, E lie on opposite sides of the chord through C, D , and IV.16 opposite-side leads to an exterior-angle contradiction.

The proposition-specific helper closing this branch is historically the same-ray IV.19 secant lemma; its general uniqueness ingredient now lives in the interface.

9 Why Same-Secant Equality Forces Coincidence

The theorem

```
hilbert_equal_angles_same_secant_unique
```

deserves separate comment.

Its input is:

$$O, B, B'$$

on one ray and a point C off that carrier, together with

$$\angle OBC \cong \angle OB'C.$$

If $B \neq B'$, one of the two points lies strictly between the origin and the other. The assumed angle congruence then identifies an exterior angle of the triangle determined by the nearer point with a remote interior angle.

Euclid's exterior-angle theorem rules this out.

Thus

$$B = B'.$$

This is a general synthetic rigidity fact, not something peculiar to circles, which is why the production refactor correctly places it in `HilbertInterfaceIV`.

10 Opposite-Ray Secant Branch

Assume now

$$B - C - E.$$

Because C lies on the chord carrier CD , the segment BE crosses the chord at C . Hence B, E lie on opposite sides of chord CD .

Since A, B are on the same side of the chord,

$$A, E$$

are on opposite sides.

The reusable production helper

```
hilbert_oppositeRay_gives_oppositeSide
```

packages exactly this order-to-half-plane conversion.

Apply IV.16 opposite-side to the circle points C, D, A, E . It produces a supplement of the angle at E congruent to

$$\angle CAD.$$

Using the IV.19 hypothesis

$$\angle CAD \cong \angle CBD$$

and the collinearity/ray relation created by $B - C - E$, the proof transports the angle at B into the corresponding remote interior angle of triangle EDB .

The result is that a remote interior angle of triangle EDB is congruent to an exterior angle at E .

Hilbert's exterior-angle theorem forbids this.

Therefore the opposite-ray secant branch is impossible.

In the original staged proof this was the theorem

```
hilbert_forder_IV19_secant_oppositeRay_impossible_stage5
```

The production module retains the proposition-specific contradiction while using semantic interface names for the general order lemmas.

11 Tangent Branch

The only remaining alternative from the circle-line dichotomy is that the line CB is tangent to the constructed circle at C .

This branch is also impossible.

Extend BA through A to a point Z :

$$B - A - Z$$

in production naming, equivalently the source extension of the line through the noncircle point and the relevant vertex.

The important side relation is that the extension point Z lies on the opposite side of chord CD from B , hence also on the opposite side from A , because A, B are same-side.

Now IV.18 applies to the tangent at C .

The tangent-chord theorem gives an angle congruence between the tangent angle at the circle point and the inscribed angle standing on the same chord.

Combining it with the IV.19 hypothesis identifies that tangent angle with the angle at the noncircle point.

But the extension point makes the tangent angle an exterior angle of a proper triangle, while the other angle is a remote interior angle.

This contradicts the exterior-angle theorem.

Hence the tangent branch is impossible.

The proposition-specific helper is

```
hilbert_forder_IV19_tangent_impossible_stage5
```

and its only numbered circle input is IV.18.

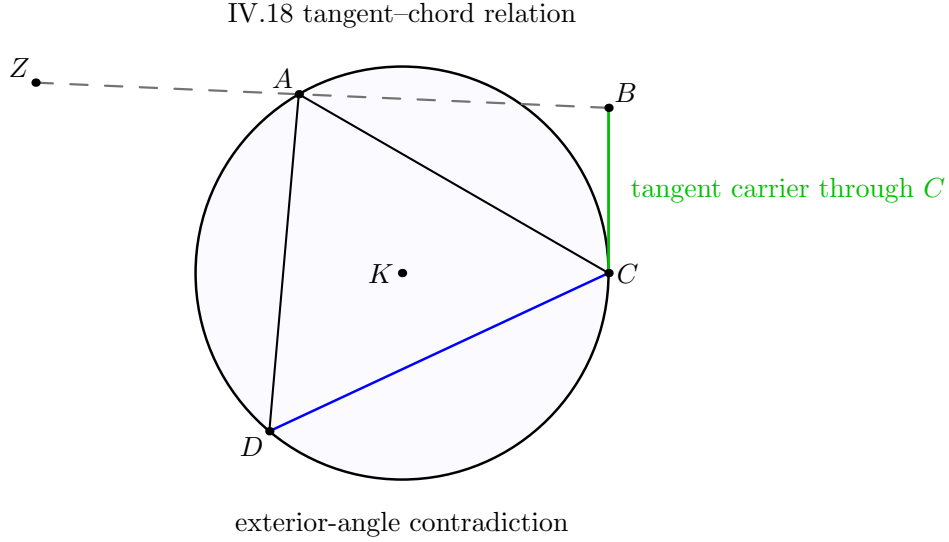


Figure 4: The tangent alternative in IV.19. If the carrier through the candidate point were tangent to the constructed circle, IV.18 would turn the tangent–chord angle into the same inscribed angle appearing in the hypothesis. After transport, an exterior angle of the relevant triangle would be congruent to a remote interior angle.

12 Completion of the Converse

The complete proof is now exhaustive.

Construct the circumcircle through C, D, A .

If B lies on it, return concyclicity.

Otherwise consider line CB .

If it is secant, let E be the second circle point.

- same ray: equal-angle secant rigidity forces $E = B$, contradicting that B is off the circle;
- opposite ray: IV.16 opposite-side creates an impossible exterior-angle congruence.

If the line is tangent, IV.18 creates another impossible exterior-angle congruence.

Every way for B to remain off the circle has therefore been eliminated.

Hence

A, C, D, B are concyclic.

The staged complete theorem was

```
hilbert_forder_IV19_circle_converse_stage6
```

The production module exports the same mathematics under the public names

```
hilbert_IV19_circle_converse_of_IV16_IV18
hilbert_IV19_circle_converse
```

13 Dependency-Explicit and Closed Theorems

IV.19 is unusual in this block because the production module exposes two useful endpoints.

The dependency-explicit theorem

```
hilbert_IV19_circle_converse_of_IV16_IV18
```

makes the proof parameters visible: it consumes the same-side and opposite-side forms of IV.16 together with IV.18.

The closed theorem

```
hilbert_IV19_circle_converse
```

supplies the production results

```
hilbert_IV16_same_side  
hilbert_IV16_opposite_side  
hilbert_IV18_tangent_chord
```

and returns the proposition interface with no theorem parameters.

This dual API is useful downstream. The explicit form records the exact logical reduction; the closed form is the convenient theorem imported by later numbered propositions.

14 Lean Formalization

The public API is:

```
HilbertForderIV19CircleConverse  
hilbert_IV19_circle_converse_of_IV16_IV18  
hilbert_IV19_circle_converse
```

The proposition-specific proof architecture contains the secant and tangent contradiction lemmas.

The general reusable ingredients have production semantic names in `HilbertInterfaceIV`:

```
hilbert_circle_line_second_or_tangent  
hilbert_equal_angles_same_secant_unique  
hilbert_secant_ray_order_split  
hilbert_oppositeRay_gives_oppositeSide
```

This ownership is important. These theorems were discovered during the IV.19 development, but none of their statements is specific to the conclusion of IV.19.

They therefore belong to the reusable interface, while `Proposition4_19.lean` owns only the proof that the secant and tangent alternatives contradict the IV.19 hypothesis.

15 Relation to IV.16

Both halves of IV.16 occur.

In the same-ray secant branch, the relevant circle points are on the same side of the chord, so IV.16 same-side gives a direct angle congruence.

In the opposite-ray branch, the line order forces the second circle point onto the opposite side of the chord, so IV.16 opposite-side gives a supplement relation.

Thus the two-way public split of IV.16 again aligns exactly with the geometry of a later proof.

16 Relation to IV.18

IV.18 is used only in the tangent branch.

This is conceptually clean:

secant branches $\longleftrightarrow$ IV.16,

whereas

tangent branch $\longleftrightarrow$ IV.18.

The circle-line dichotomy therefore routes the proof directly to the theorem appropriate for each geometric type.

17 Circumcircle Existence and the Converse Direction

The converse nature of IV.19 makes circumcircle existence foundationally visible.

The forward same-segment theorem IV.16 begins with an already given circle. IV.19 does not.

It must first construct a circle through the three noncollinear points C, D, A .

Only after that construction can the fourth point B be tested against the circle.

This is an important logical distinction:

equal angles $\nRightarrow$ “there is already a circle”.

The proof must use the established circumcenter existence theorem to create the candidate circle.

18 Hilbert Components Used

18.1 Circumcenter existence

The triangle CDA is proper because A is off chord CD . The Euclidean-plane circumcenter theorem supplies the centre K .

18.2 Circle-line dichotomy

A line through a noncentral circle point is either secant again or tangent. This theorem supplies the exhaustive top-level classification once B is assumed off the candidate circle.

18.3 Ray order and plane separation

The secant point E must be classified relative to ray CB . The order determines whether the relevant pair of points is same-side or opposite-side with respect to chord CD .

18.4 Angle rigidity on a secant

The same-ray branch closes through `hilbert_equal_angles_same_secant_unique`. Its proof is ultimately an exterior-angle contradiction.

18.5 IV.16 and IV.18

IV.16 controls the secant branches; IV.18 controls the tangent branch.

18.6 Exterior-angle comparison

Every impossible branch eventually reduces to the same geometric principle: an exterior angle of a proper triangle cannot be congruent to the appropriate remote interior angle.

This repeated terminal contradiction is the synthetic substitute for the usual informal uniqueness intuition.

19 Foundational Boundary

IV.19 introduces no new circle axiom.

It does use the existing theorem guaranteeing a circumcircle for a proper triangle. Once that circle exists, the rest of the proof uses:

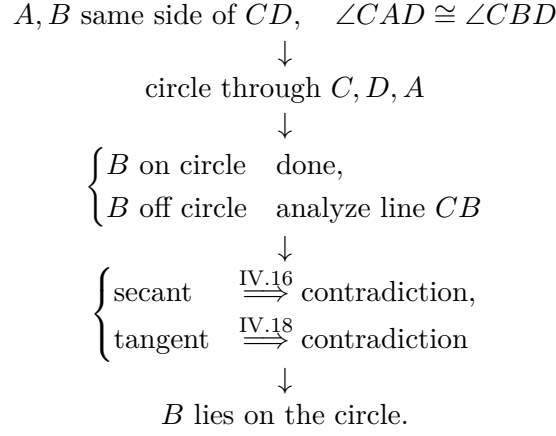
- incidence and line uniqueness;
- ray order;
- plane separation;
- the secant/tangent dichotomy;
- IV.16 same-side and opposite-side;
- IV.18;
- synthetic exterior-angle comparison.

No coordinate equation of a circle, no determinant criterion for concyclicity, and no numerical angle measure appear.

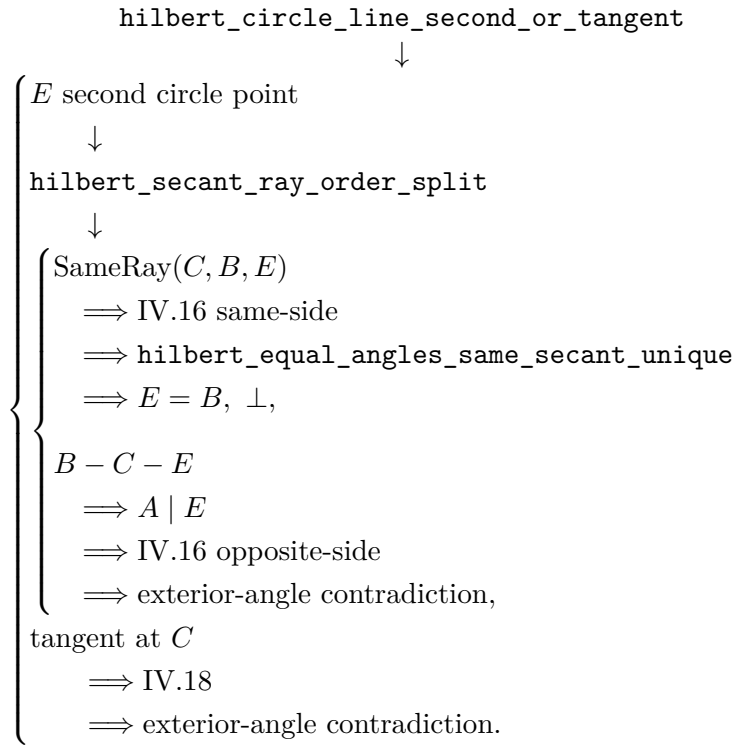
The converse is therefore proved entirely within the synthetic Hilbert framework.

20 Classical and Formal Dependency Structures

Former's proof can be summarized as:



The formal proof expands the secant line order:



The production import graph is:

```

HilbertInterfaceIV
Proposition4_16
Proposition4_18
-> Proposition4_19

```

There is no direct dependency on IV.17 because IV.18 already encapsulates the tangent-chord theorem needed here.

21 Architectural Significance

IV.19 is where the semantic extraction of general helpers from the old monolithic development becomes especially important.

The following results were first needed while proving IV.19:

```
hilbert_circle_line_second_or_tangent
hilbert_equal_angles_same_secant_unique
hilbert_secant_ray_order_split
hilbert_oppositeRay_gives_oppositeSide
```

But none is mathematically “Proposition IV.19”.

The production refactor therefore places them in `HilbertInterfaceIV`.

That move has two consequences:

1. IV.17 can reuse the secant-order split without depending forward on IV.19;
2. IV.19 itself becomes a numbered theorem module containing only its proposition-specific converse argument.

This is one of the clearest examples in the Book IV block of semantic module ownership replacing historical proof-development ownership.

22 Diagrammatic Audit

Four figures are appropriate.

1. The source same-segment converse configuration.
2. The constructed circumcircle through three points with B not yet known to lie on it.
3. A two-panel figure for the two secant ray orders.
4. The tangent contradiction controlled by IV.18.

These figures mirror the actual logical branching and are preferable to a single crowded picture containing all auxiliary points simultaneously.

23 Conclusion

Forder IV.19 is the first full circle-converse theorem in this sequence.

The proof does not infer concyclicity from an angle equation by a black-box criterion. It constructs the circumcircle through three points and then eliminates every possible way for the fourth point to stay off that circle.

If the line through the fourth point is secant, IV.16 and secant rigidity eliminate both possible ray orders. If it is tangent, IV.18 and the exterior-angle theorem give the contradiction.

The proposition also illustrates the production architecture particularly well. General secant/tangent and ray-order facts live in `HilbertInterfaceIV`; the numbered module owns only the circle converse itself and its secant/tangent contradiction branches.

The result is a source-faithful synthetic reconstruction of the same-segment converse, with the hidden existence, order, and uniqueness arguments made fully explicit.

References

- [1] Henry George Forder, *Foundations of Euclidean Geometry*, Cambridge University Press, Cambridge, 1927.